



3162 - Does star formation require molecular gas?

Cycle: 2, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	IZW18-POS1	MIRI Medium Resolution Spectroscopy	(1) IZW18-POS1
	2	IZW18-OFF	MIRI Medium Resolution Spectroscopy	(4) IZW18-OFFSET-BG
	3	IZW18-POS2	MIRI Medium Resolution Spectroscopy	(2) IZW18-POS2

ABSTRACT

We propose to carry out a strong test of the hypothesis that star formation requires the presence of molecular gas. To do this, we will search for H₂ rotational line emission from the extremely low metallicity star-forming dwarf galaxy I Zw 18 using the MIRI IFU. Successful detection of this emission will allow us to constrain the molecular gas mass in I Zw 18, enabling us to compare it with the predictions of the H₂-regulated star formation model. Alternatively, non-detection of the lines will allow us to rule out the presence of any significant mass of H₂, providing strong support for models in which the presence of H₂ is not necessary for star formation.

In addition, because we require observations in all three grating settings to cover the H₂ lines, our proposed observations will yield deep MIR spectra of I Zw 18 covering a broad range of wavelengths and including a number of nebular emission lines and PAH features. These observations will therefore enable a very broad range of additional science.

OBSERVING DESCRIPTION

We will observe I Zw 18 using the MIRI IFU. We will take science observations towards two pointings, one centred on I Zw 18 NW and the other on I Zw 18 SE. To allow for background subtraction, we will also observe an off position located far away from the galaxy.

Each pointing will involve observations with all three MIRI IFU sub-bands. During the MIRI IFU observations, we will obtain simultaneous observations with the MIRI imager, using the F560W, F1130W and F2100W filters. We adopt a standard 4-point dither pattern towards each pointing, with 1 exposure per dither position and 2 integrations per exposure. Read-out of the IFU will be done using the SLOWR1 pattern, in order to avoid hitting the data volume limit. Read-out of the simultaneous imaging will use the FASTR1 pattern.

We utilize POS-TARGS to perform offsets to a clean background field in detector-FOV coordinates. This allows us to offset in such a direction so as to align the MIRI imaging FOV on the target itself while the background IFU data are being collected. Doing so will allow us to obtain high quality, broad continuum and PAH (5.6, 11.3, and 21 micron) maps in MIRI simultaneous imaging, greatly increasing our ability to sample the changing starlight heating environment.

Proposal 3162 - Targets - Does star formation require molecular gas?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	IZW18-POS1	RA: 09 34 2.3338 (143.5097242d) Dec: +55 14 24.00 (55.24000d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Blue compact dwarf galaxies]				
	(2)	IZW18-POS2	RA: 09 34 1.9773 (143.5082388d) Dec: +55 14 28.30 (55.24119d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Blue compact dwarf galaxies]				
	(4)	IZW18-OFFSET-BG	RA: 09 34 1.9773 (143.5082388d) Dec: +55 14 28.30 (55.24119d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Blue compact dwarf galaxies]				

Proposal 3162 - Observation 1 - Does star formation require molecular gas?

Observation	Proposal 3162, Observation 1: IZW18-POS1												Mon Jul 10 21:02:32 GMT 2023	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[IZW18-OFF (Obs 2)]													
Diagnostics	(IZW18-POS1 (Obs 1)) Warning (Form): Imager Filter overlap.													
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(1)	IZW18-POS1	RA: 09 34 2.3338 (143.5097242d)					Epoch of Position: 2015.5						
			Dec: +55 14 24.00 (55.24000d)											
			Equinox: J2000											
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
Acquisition	Category=Galaxy													
	Description=[Blue compact dwarf galaxies]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
	FND	ALL				YES			FULL			NEUTRAL		
Dithers	#	Dither Type					Optimized For				Direction			
	1	4-Point					EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F2100W	FASTR1	5	2	1	Dither 1	4	8	122.102		
	1	LONG(C)	MRSLONG		SLOWR1	19	2	1	Dither 1	4	8	3726.828		
	1	LONG(C)	MRSSHORT		SLOWR1	19	2	1	Dither 1	4	8	3726.828		
	2		IMAGER	F1130W	FASTR1	5	2	1	Dither 1	4	8	122.102		
	2	MEDIUM(B)	MRSLONG		SLOWR1	18	2	1	Dither 1	4	8	3535.708		
	2	MEDIUM(B)	MRSSHORT		SLOWR1	18	2	1	Dither 1	4	8	3535.708		
	3		IMAGER	F560W	FASTR1	5	2	1	Dither 1	4	8	122.102		
	3	SHORT(A)	MRSLONG		SLOWR1	15	2	1	Dither 1	4	8	2962.35		
	3	SHORT(A)	MRSSHORT		SLOWR1	15	2	1	Dither 1	4	8	2962.35		

Proposal 3162 - Observation 1 - Does star formation require molecular gas?

Special Requirements	No Parallel Attachments Sequence Observations 1, 2, 3, Non-interruptible
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Proposal 3162 - Observation 2 - Does star formation require molecular gas?

Observation	Proposal 3162, Observation 2: IZW18-OFF												Mon Jul 10 21:02:32 GMT 2023
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [IZW18-POS1 (Obs 1), IZW18-POS2 (Obs 3)]												
Diagnostics	(IZW18-OFF (Obs 2)) Warning (Form): Imager Filter overlap.												
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(4)	IZW18-OFFSET-BG	RA: 09 34 1.9773 (143.5082388d)				Epoch of Position: 2015.5						
			Dec: +55 14 28.30 (55.24119d)										
			Equinox: J2000										
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Galaxy												
	Description=[Blue compact dwarf galaxies]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction	
	FND	ALL				YES				FULL		NEUTRAL	
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F2100W	FASTR1	20	2	1	Dither 1	4	8	455.107	
	1	LONG(C)	MRSLONG		SLOWR1	19	2	1	Dither 1	4	8	3726.828	
	1	LONG(C)	MRSSHORT		SLOWR1	19	2	1	Dither 1	4	8	3726.828	
	2		IMAGER	F1130W	FASTR1	110	2	1	Dither 1	4	8	2453.135	
	2	MEDIUM(B)	MRSLONG		SLOWR1	18	2	1	Dither 1	4	8	3535.708	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	18	2	1	Dither 1	4	8	3535.708	
	3		IMAGER	F560W	FASTR1	110	2	1	Dither 1	4	8	2453.135	
	3	SHORT(A)	MRSLONG		SLOWR1	15	2	1	Dither 1	4	8	2962.35	
	3	SHORT(A)	MRSSHORT		SLOWR1	15	2	1	Dither 1	4	8	2962.35	

Proposal 3162 - Observation 2 - Does star formation require molecular gas?

Special Requirements	Offset -53.04611792029557 arcsec, -52.80858730034301 arcsec No Parallel Attachments Sequence Observations 1, 2, 3, Non-interruptible
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Proposal 3162 - Observation 3 - Does star formation require molecular gas?

Observation	Proposal 3162, Observation 3: IZW18-POS2												Mon Jul 10 21:02:32 GMT 2023
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[IZW18-OFF (Obs 2)]												
Diagnostics	(IZW18-POS2 (Obs 3)) Warning (Form): Imager Filter overlap.												
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(2)	IZW18-POS2	RA: 09 34 1.9773 (143.5082388d)					Epoch of Position: 2015.5					
			Dec: +55 14 28.30 (55.24119d)										
			Equinox: J2000										
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Blue compact dwarf galaxies]											
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	FND	ALL				YES			FULL		NEUTRAL		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F2100W	FASTR1	5	2	1	Dither 1	4	8	122.102	
	1	LONG(C)	MRSLONG		SLOWR1	19	2	1	Dither 1	4	8	3726.828	
	1	LONG(C)	MRSSHORT		SLOWR1	19	2	1	Dither 1	4	8	3726.828	
	2		IMAGER	F1130W	FASTR1	5	2	1	Dither 1	4	8	122.102	
	2	MEDIUM(B)	MRSLONG		SLOWR1	18	2	1	Dither 1	4	8	3535.708	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	18	2	1	Dither 1	4	8	3535.708	
	3		IMAGER	F560W	FASTR1	5	2	1	Dither 1	4	8	122.102	
	3	SHORT(A)	MRSLONG		SLOWR1	15	2	1	Dither 1	4	8	2962.35	
	3	SHORT(A)	MRSSHORT		SLOWR1	15	2	1	Dither 1	4	8	2962.35	

Proposal 3162 - Observation 3 - Does star formation require molecular gas?

Special Requirements	No Parallel Attachments Sequence Observations 1, 2, 3, Non-interruptible
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